# Siddharth R. Chintalapalli

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## Education

**Georgia Institute of Technology | Atlanta, GA** *Aug 2026 – Present*
Bachelor of Science in Electrical Engineering, GPA: 4.0/4.0 Graduation: May 2029
Sophomore/Rising Junior in Fall 26; Junior in Summer/Fall 27

## Skills

**Programming:** Python [Proficient], Arduino [Expert], C++ [Proficient], BetaFlight (Drone Programming) [Expert], Git [Proficient], GitFlow [Proficient], GitHub [Proficient]
**Hardware:** Arduino [Expert], Laser Engraving [Expert], 3D Printing [Proficient]
**Software:** Altium Designer [Proficient] TinkerCAD [Expert], Computer-Aided Design: Solidworks, Fusion 360, AutoCAD [Proficient], BambuLabs [Proficient], Adobe Illustrator [Proficient], Computational Fluid Dynamics (CFD) [Proficient], Schematics [Expert]
**Certifications:** Federal Aviation Administration Part 107 License
**Knowledge:** Digital Logic Design [Expert], CMOS networks [Expert], Gate Level Logic [Expert], Sequential Circuits [Proficient]
**Communication:** Microsoft 365 [Expert], Design Proposals [Expert], Technical Reports [Expert]
**Languages:** English [Fluent], Telugu [Proficient], Spanish [Proficient]

## Experience

**Georgia Institute of Technology HyTech Racing | Electrical Subsystem Team Member** *Aug 2026 – Present*
- Work on HT11 FSAE Electric Racecar's electrical sub-team that develops low & high voltage systems and firmware of racecar.
- Work on onboarding PCB Design via Altium and firmware design utilizing PlatformIO, and C++ to collect temperature data.
- Designing and testing ethernet implementation on rev 9 vehicle dashboard for faster data transfer and high visual processing.
- Developing Charging Control Unit's Unit Test Code to produce virtual test environment for code prior to flashing to hardware.

**Innovation Academy Motorsports | *Founder, Team Lead (Specialized Electrical Systems)*** *Jan 2025 – May 2026*
- Founded motorsports organization that competed in high school racing divisions for the 1st time & placed 9th at state-level competition & 1st in local intraschool competition.
- Specialized in electrical systems and circuitry of electric race cars, computer-aided design & CFD analysis of chassis.
- Built 2 custom-aluminum chassis electric racecars, designed battery system, driver controls, and circuit wiring for racecars.
- Operated in chassis manufacturing and welding, drivetrain systems, & body paneling.

**PE Structures & Associates, LLC | Roswell, GA | *Engineering Intern*** *Feb – Aug 2025*
- Worked on 3 Public General Contracting Engineering Projects related to riverbank restoration and park development.
- Developed office floor plans via AutoCAD & analyzed construction drawings via Bluebeam Revu.
- Worked on-site with photography, recording & safety management & led efforts to integrate drone aerial footage & technology.

## Personal Projects

**First Person View Drones & Drone Gimbal Innovation** *Jan – April 2025*
- Engineered and programmed two FPV drones from scratch utilizing Betaflight as the primary program language.
- Innovated a 3D-printed multipurpose video gimbal via CAD & Bambu Labs; programmed the gimbal controller via Simple BGC.

**Arduino Video Game Controller** *Jan - Mar 2025*
- Fabricated custom video game controller, programmed Arduino Nano to execute game operations, integrated a joystick & four buttons & soldered electrical components for functional hardware integration.
- Prototyped casing via iterative CAD design and modification, along with 3D printing via Bambu Labs & Bambu X1 Carbon.

**$CO_2$ Dragster Racecar** *Nov 2024 - Mar 2026*
- Designed aerodynamic, light-weight Shell Dragster devised for max speed via CAD and ran CFD sims to improve aerodynamics
- Manufactured prototypes via woodworking & 3D printing, testing different filaments and settings via Bambu Labs

## Leadership

**Innovation Academy Foundation STEM Camp | *Founder, Engineering Student Lead*** *May 2025 – June 2026*
- Founded and executed two successful IAF STEM Summer Camp with over 200 middle school campers
- Raised $50,000+ in revenue directed towards funding school clubs and organizations for following year
- Instructed campers on engaging STEM sessions such as, Model Rockets and Battle Bots, to teach core engineering concepts

**Innovation Academy Technology Student Association | *Chapter Officer, Competitor*** *Aug 2024 – May 2026*
- Led chapter to ranking 5/244 at state level (21-rank improvement) & increased club membership by 100+ students.
- Won Top 10 in 6 engineering & STEM events at the State Level – Electrathon EV Rally Race (9th), CO2 Dragster (8th), Drone UAV Challenge (1st & 3rd), Structural Design & Engineering (10th), Website Design (6th).
- Restructured chapter program: introduced after-school practice, competition prep sessions & developed 1st intraschool comp.